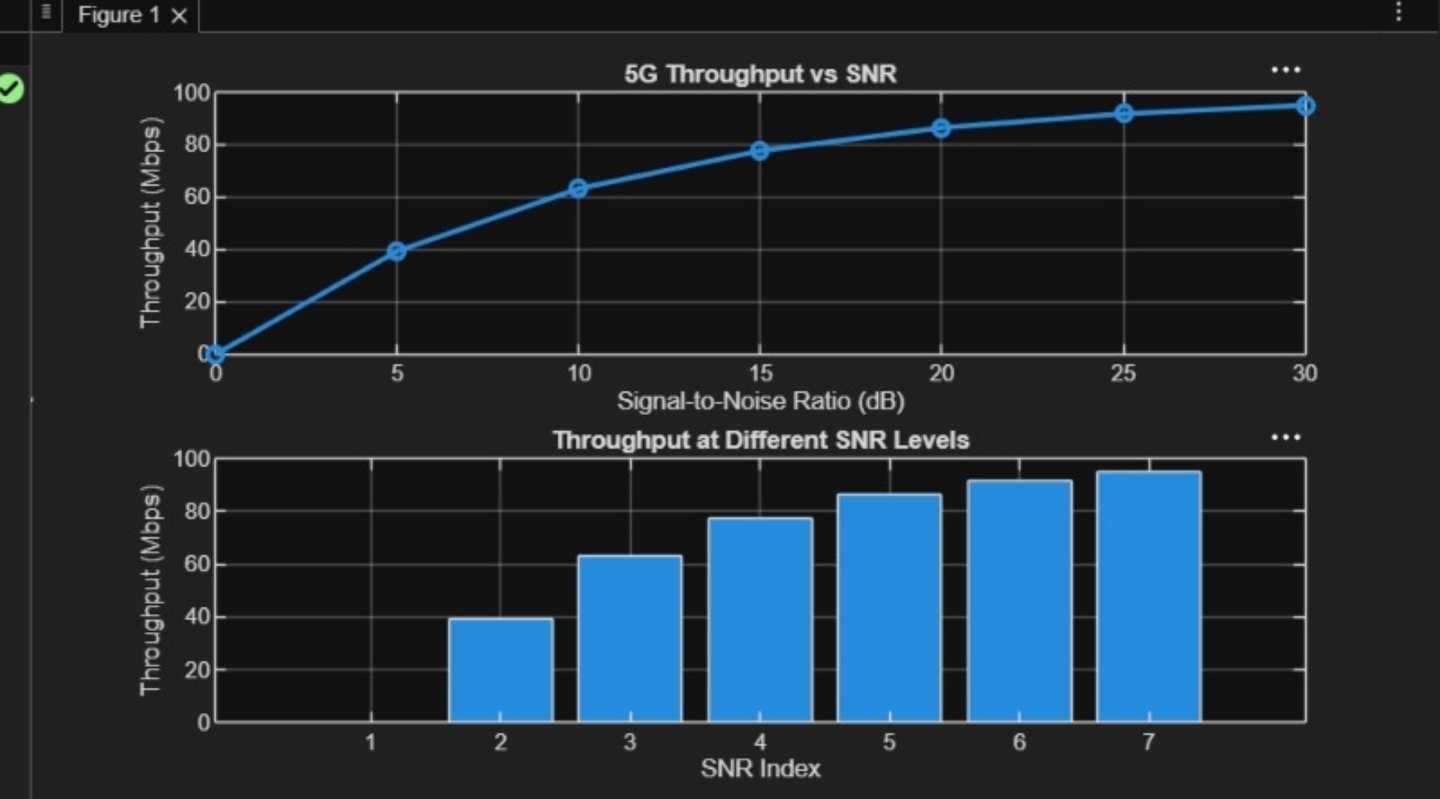


Save As, Copy Figure, Print, Show Code, FILE, LABELS AND ANNOTATIONS: Title, Subtitle, X-Label, Y-Label, Z-Label, Legend, Colorbar, Grid, X-Grid, Y-Grid

```
untitled36.m x untitled37.m x untitled38.m x untitled39.m x untitled40.m x
/MATLAB Drive/untitled40.m
3 close all;
4 % SNR values (dB)
5 SNR = 0:5:30;
6 % Maximum Throughput (Mbps)
7 Max_TP = 100;
8 % Throughput calculation
9 Throughput = Max_TP*(1-exp(-SNR/10));
10 disp('SNR(dB) Throughput(Mbps)')
11 disp([SNR' Throughput'])
12 figure
13 % ----- Graph 1 -----
14 subplot(2,1,1)
15 plot(SNR,Throughput,'-o','LineWidth',2)
16 title('5G Throughput vs SNR')
17 xlabel('Signal-to-Noise Ratio (dB)')
18 ylabel('Throughput (Mbps)')
19 grid on
20 % ----- Graph 2 -----
21 subplot(2,1,2)
22 bar(Throughput)
23 title('Throughput at Different SNR Levels')
24 xlabel('SNR Index')
25 ylabel('Throughput (Mbps)')
```



Command Window

|         |         |
|---------|---------|
| 10.0000 | 63.2121 |
| 15.0000 | 77.6870 |
| 20.0000 | 86.4665 |
| 25.0000 | 91.7915 |
| 30.0000 | 95.0213 |

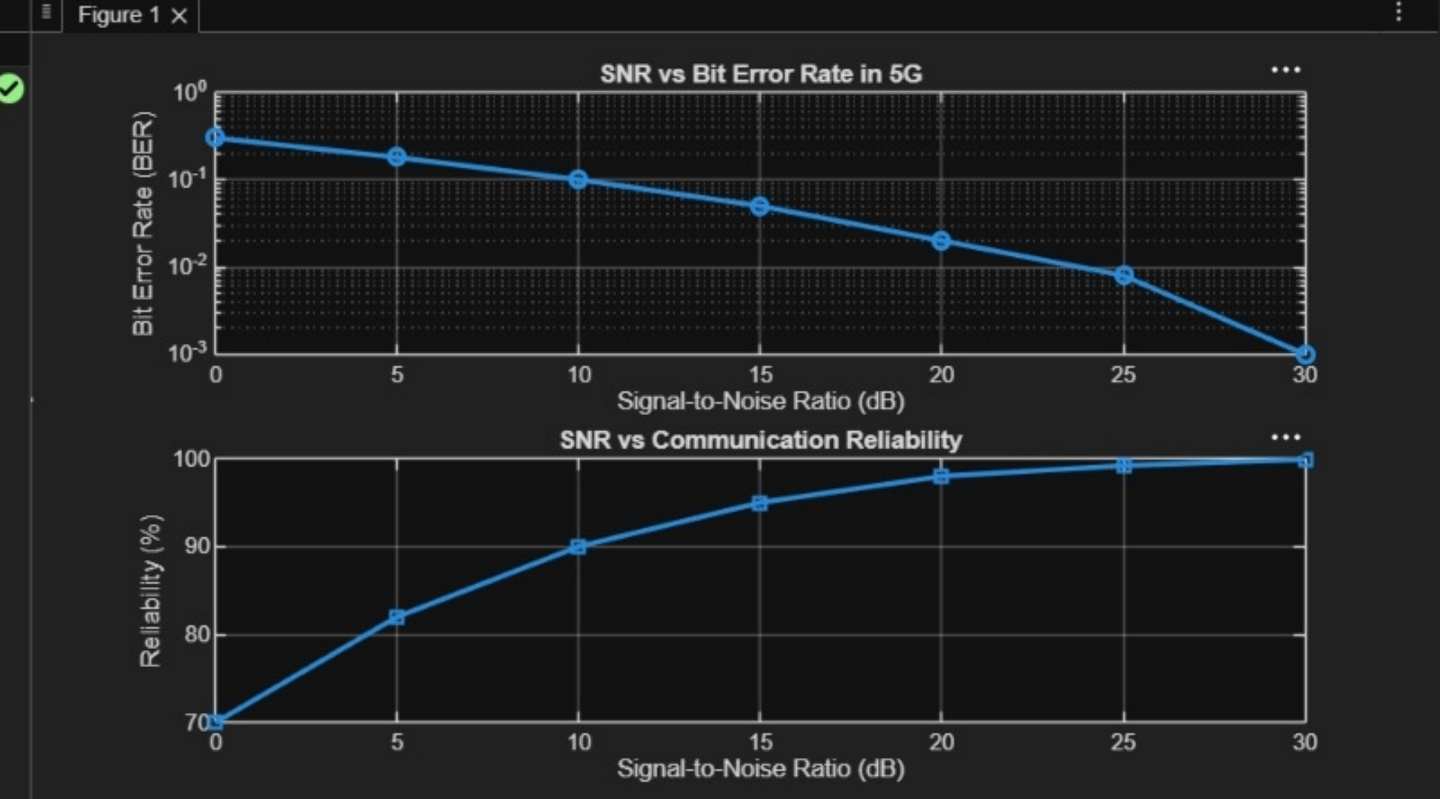
>>

Save As, Copy Figure, Print, Show Code, FILE

TITLE, Subtitle, X-Label, Y-Label, Z-Label, Legend, Colorbar, Grid, X-Grid, Y-Grid

LABELS AND ANNOTATIONS

```
untitled36.m x untitled37.m x untitled38.m x untitled39.m x untitled40.m x untitled41.m x +
/MATLAB Drive/untitled41.m
3 close all;
4 % SNR values (dB)
5 SNR = 0:5:30;
6 % Simulated Bit Error Rate (BER)
7 BER = [0.30 0.18 0.10 0.05 0.02 0.008 0.001];
8 % Reliability (%)
9 Reliability = (1-BER)*100;
10 disp('SNR(dB) BER Reliability(%)')
11 disp([SNR' BER' Reliability'])
12 figure
13 % ----- Graph 1 -----
14 subplot(2,1,1)
15 semilogy(SNR,BER,'-o','LineWidth',2)
16 title('SNR vs Bit Error Rate in 5G')
17 xlabel('Signal-to-Noise Ratio (dB)')
18 ylabel('Bit Error Rate (BER)')
19 grid on
20 % ----- Graph 2 -----
21 subplot(2,1,2)
22 plot(SNR,Reliability,'-s','LineWidth',2)
23 title('SNR vs Communication Reliability')
24 xlabel('Signal-to-Noise Ratio (dB)')
25 ylabel('Reliability (%)')
```



Command Window

|         |        |         |
|---------|--------|---------|
| 10.0000 | 0.1000 | 90.0000 |
| 15.0000 | 0.0500 | 95.0000 |
| 20.0000 | 0.0200 | 98.0000 |
| 25.0000 | 0.0080 | 99.2000 |
| 30.0000 | 0.0010 | 99.9000 |

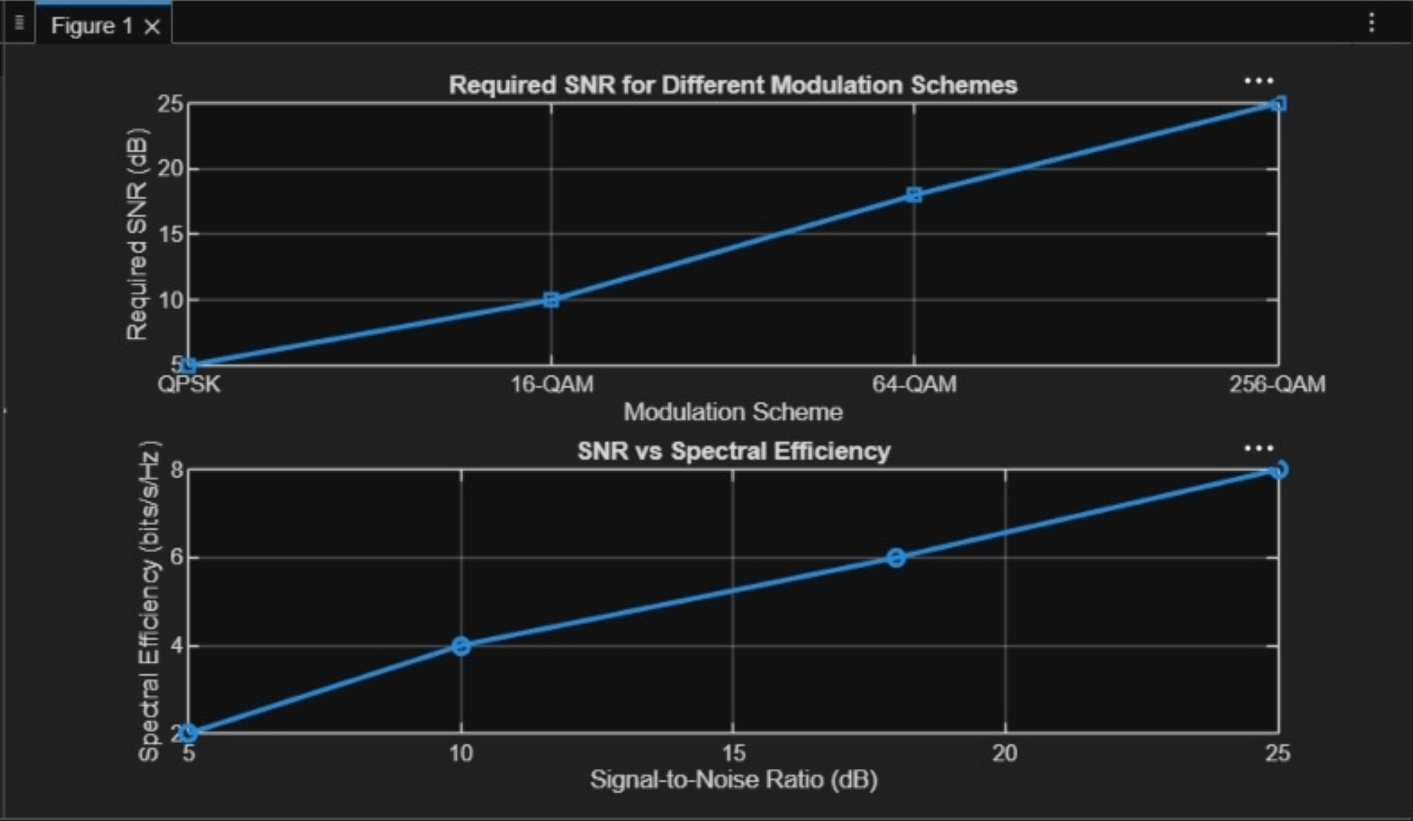
>>

Save As, Copy Figure, Print, Show Code, FILE

TITLE, SUBTITLE, X-LABEL, Y-LABEL, Z-LABEL, LEGEND, COLORBAR, GRID, X-GRID, Y-GRID

LABELS AND ANNOTATIONS

```
untitled36.m x untitled37.m x untitled38.m x untitled39.m x untitled40.m x untitled41.m x untitled42.m x
/MATLAB Drive/untitled42.m
6 % Required SNR (dB)
7 SNR = [5 10 18 25];
8 % Spectral Efficiency (bits/s/Hz)
9 Efficiency = [2 4 6 8];
10 disp('Required SNR (dB) and Spectral Efficiency')
11 disp(table(Modulation,'SNR','Efficiency', ...
12 'VariableNames',{'Modulation','SNR_dB','Efficiency'}))
13 figure
14 % ----- Graph 1 -----
15 subplot(2,1,2)
16 plot(SNR,Efficiency,'-o','LineWidth',2)
17 title('SNR vs Spectral Efficiency')
18 xlabel('Signal-to-Noise Ratio (dB)')
19 ylabel('Spectral Efficiency (bits/s/Hz)')
20 grid on
21 % ----- Graph 2 -----
22 subplot(2,1,1)
23 plot(1:4,SNR,'-s','LineWidth',2)
24 xticks(1:4)
25 xticklabels(Modulation)
26 title('Required SNR for Different Modulation Schemes')
27 xlabel('Modulation Scheme')
28 ylabel('Required SNR (dB)')
```



Command Window

|               |    |   |
|---------------|----|---|
| { 'QPSK' }    | 5  | 2 |
| { '16-QAM' }  | 10 | 4 |
| { '64-QAM' }  | 18 | 6 |
| { '256-QAM' } | 25 | 8 |

>>